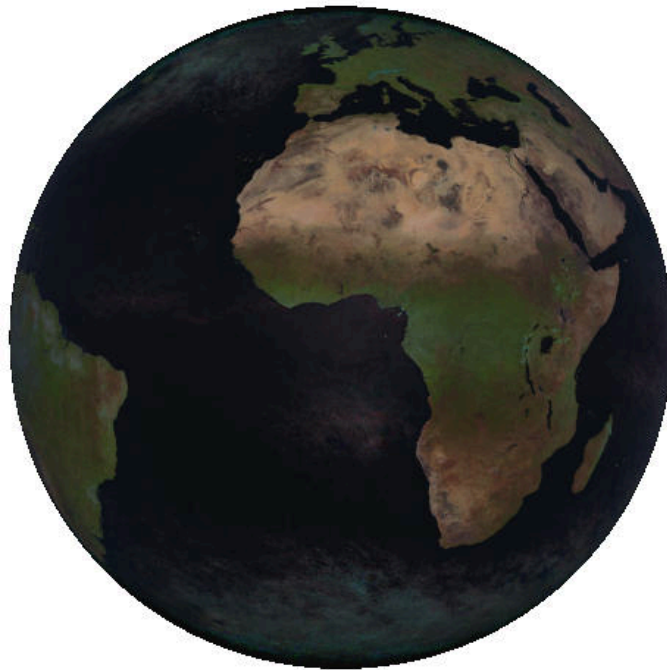


Global Weather Monitoring: Report on the Phenological Properties of African Vegetation

Ben C Potter
School of Physics and Astronomy
University of Nottingham

September 13, 2026



Abstract

In this report, we investigated some of the African continent's phenological characteristics. It includes an investigation into comparing total vegetation coverage to total cloud coverage, as well as looking into total vegetation coverage on its own, and the position and spread of the vegetation. Through using a method to generating cloud-free images for every 14 days, and analysing NDVI data from those, we investigated these areas. We found the lead times between the peaks of vegetation coverage and cloud coverage to be 2 ± 4 weeks. The cloud coverage has a main frequency of 1.0 ± 0.2 per year. The total vegetation coverage has a main frequency of 1.9 ± 0.2 per year, owing to its double-dip nature. The position and spread of the vegetation has a position frequency of 0.9 ± 0.1 per year, and the spread is larger when the vegetation is in the north than when it is in the south. The highest total vegetation coverage occurs in the spring and autumn when the vegetation moves north to south and vice-versa.

Contents

1	Introduction	3
2	Methodology	5
2.1	Handling the Images	5
2.2	Cloud Free Image	7
2.3	Normalised Difference Vegetation Index	9
3	Results	13
4	Discussion of Results	16
5	Conclusions	23



1 Introduction

With the launches of Sputnik from the USSR and the Explorer program in the USA, research and design began on specifically meteorological satellites. These early satellites held television cameras and radiometers, mainly in order to pursue better weather forecasting. From this came more high resolution radiometers, mainly infrared and red. The first multi spectral scanner was launched on Landsat-1, and with that NASA began to fund more remote-sensing projects, which paved the way for more and more satellites to be launched specifically for this purpose.

The series of satellites known as MeteoSat Second Generation (MSG) hold an instrument known as SEVIRI, which provides excellent high-quality images of the earth, as viewed from both $0^{\circ}\text{N } 0^{\circ}\text{W}$, and $0^{\circ}\text{N } 41.5^{\circ}\text{W}$. In this project, we use the $0^{\circ}\text{N } 0^{\circ}\text{W}$ satellite in order to capture images of Africa.

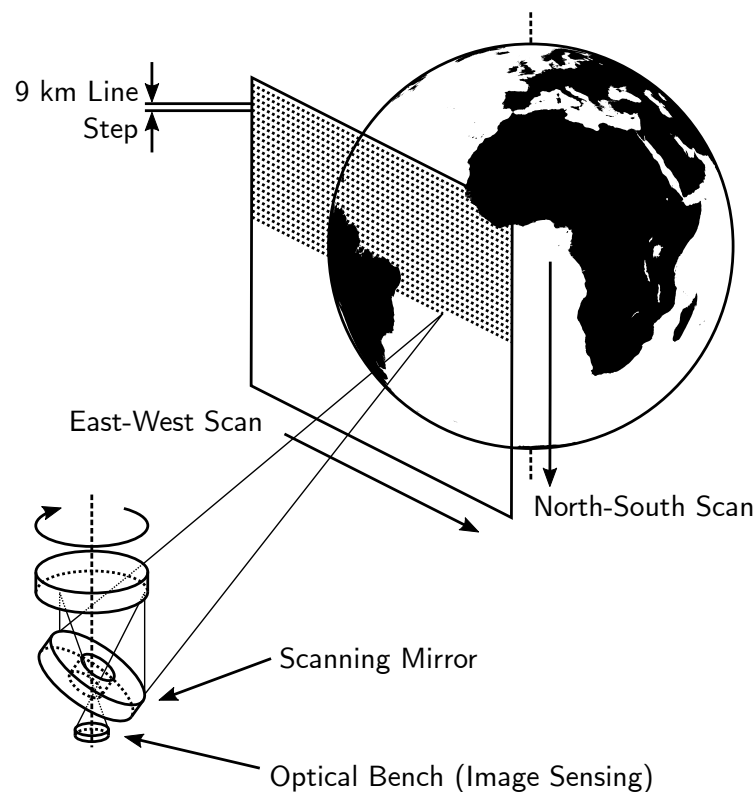


Figure 1: How the MSG SEVIRI image sensor mechanically scans the earth.

This series of satellites has 12 channels, 4 visible and near infrared channels, with 8 infrared channels. On those channels, it captures images every 15 minutes at a roughly 3km ground resolution. As we can see in Figure 1, the satellite's mirror spins within the satellite, which enables the satellite to produce an image of the whole earth. Each individual detection unit is put into the field of view through a "flip-flop" mechanism. This enables a full scan of the earth for all 12 spectral channels every 12 minutes, enabling an image to be produced every 15 minutes [Aminou, 2002].

The above table was obtained from [Weymiens and Oremus, 1999]. The first MSG satellite began operation at the start of 2004, so there is data from then until now to use. This report uses multiple images in order to create a cloud free image. An RGB image can then be created



in order to more easily observe where the "green" vegetation is, for the purposes of comparison with the later created normalised difference vegetation index (*NDVI*) map. This is used to see the visual changes occurring on the surface of the earth. This report will use 2 channels in order to find the position of vegetation across the African Continent.

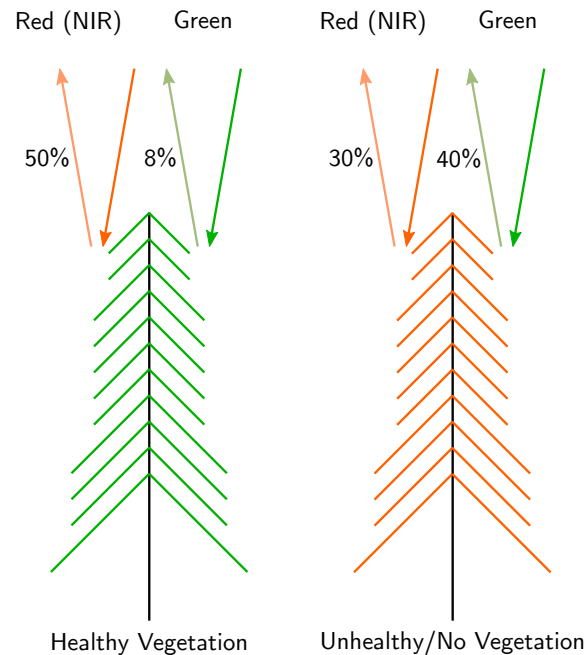


Figure 2: How the 2 channels (Red (NIR) and Green) are used to find the location of vegetation.

In Figure 2, there is a clear difference in the reflectivity between the two spectral bands, NIR (Red) and Green. Using images obtained from the MSG satellite, specifically the VIS 0.8 and IR 1.6 ¹ channels [Weymiers and Oremus, 1999], this report will use *NDVI* to locate the areas where vegetation is visible. *NDVI* was first used in a paper [Rouse Jr et al., 1973] exploring the vegetation growth in the great plains area of the United States of America. The authors of the paper created *NDVI* as a way to effectively measure the vegetation area, regardless of the brightness of the image, as they were experiencing different brightness levels due to the sun zenith angle. As the study covered the entire length of the US, from the US-Canadian border to the southern tip of Texas, the sun angle over each area was vastly different due to the variance of latitudes. We encounter a similar problem, as the African Continent covers approximately 70° of latitude, from 37°N to 34°S. Interestingly, in their paper they cover a similar aspect of vegetation growth as we do, which they refer to as vernal advancement and retrogradation, or the "green wave effect". This effect seems to be of particular importance to the migration patterns of different animals, including waterfowl [Shariatnajaabadi et al., 2014], Arctic breeding geese [Si et al., 2015], and migratory northern Ungulates [Bischof et al., 2012].

This study of cyclic and seasonal natural phenomena is known as phenology. Phenology is a centuries-old subject, largely thought to have been first started by Robert Marsham, the father of modern phenological recording, in 1736. He began to record the "Indications of Spring" on his estate in Norfolk. These were events such as flowering or the first flights of

¹The numbers indicate the spectral channel wavelength in micrometres.



insects, and these observations were continued by generations of the Marsham family up until the death of Mary Marsham in 1958 [Sparks and Carey, 1995]. By observing this for that long, it has allowed the effect of climate warming to be observed by noticing how much earlier those effects were happening. The green wave effect, fortunately for us, is much easier to see from space than the flowering of plants.

This report will show that the African content is highly phenological in it's behaviour. It will show how we obtain the images for analysis and how we handle them to be put into the program. Then it will show how they are analysed to create cloud free images, and how they are subsequently used to produce maps of the vegetation on the African landmass. It will then illustrate how the summaries of the results were produced.

2 Methodology

2.1 Handling the Images

Channel	Absorption Band	Channel Type	Nominal Wavelength (μm)	Centre Spectral Bandwidth (μm)
HRV	Visible High Resolution		0.75	0.6 to 0.9
VIS 0.6	VNIR Core Imager		0.635	0.56 to 0.71
VIS 0.8	VNIR Core Imager		0.81	0.74 to 0.88
IR 1.6	VNIR Core Imager		1.64	1.50 to 1.78
IR 3.9	IR/Window Core Imager		3.92	3.48 to 4.36
WV 6.2	Water Vapour Core Imager		6.25	5.35 to 7.15
WV 7.3	Water Vapour Pseudo-Sounding		7.35	6.85 to 7.85
IR 8.7	IR/Window Core Imager		8.70	8.30 to 9.10
IR 9.7	IR/Ozone Pseudo-Sounding		9.66	9.38 to 9.94
IR 10.8	IR/Window Core Imager		10.80	9.80 to 11.80
IR 12.0	IR/Window Core Imager		12.00	11.00 to 13.00
IR 13.4	IR/Carbon Diox. Pseudo-Sounding		13.40	12.40 to 14.40

Table 1: The various channels available for the SEVIRI instrument. Obtained from Weymiens and Oremus [1999].

In order to create a good cloud free image, we first need to choose the channels we need carefully. We have a choice of the channels from the SEVIRI instrument, shown in Table 1. In order to get a good RGB image of the ground, we need 3 channels that image the ground, and have a wavelength that corresponds roughly to one of the Red, Green or Blue. If we look at the first channel, HRV, we can discount this because it only looks at part of the earth.



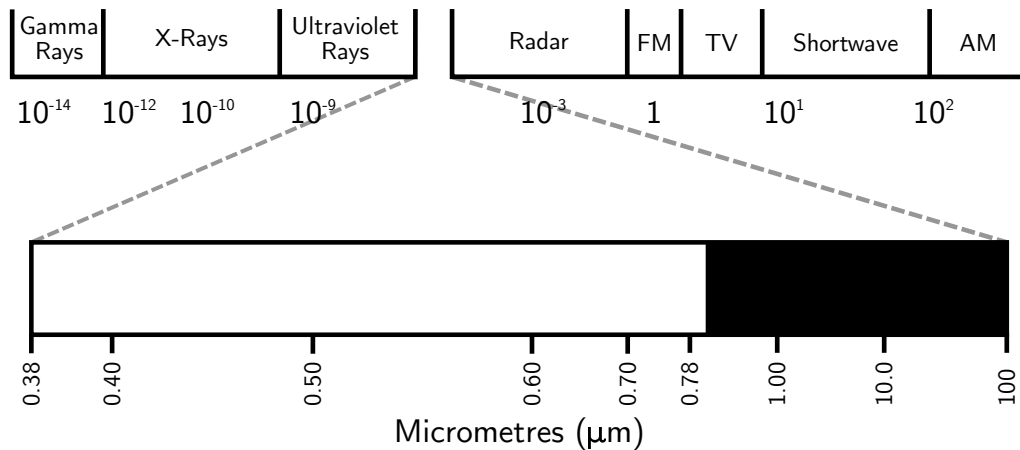


Figure 3: The light spectrum showing specifically an expanded view of where we need to choose the channels from.

The channel IR 13.4 is used for the detection of CO₂, and along with IR 12.0 has too long a wavelength to use for the red channel in the RGB image. The IR 9.7 channel is used for ozone imaging, and along with IR 10.8 and IR 3.9, also has too long a wavelength to use for any of the red, green or blue channels. WV 6.2 and WV 7.2 are both used to image water vapour in the atmosphere, and so aren't very useful for imaging the ground. We are therefore left with three channels we can use, VIS 0.6, VIS 0.8, and IR 1.6. We use the VIS 0.6 channel for blue, the VIS 0.8 for green, and the IR 1.6 channel for red. Although the VIS 0.6 channel has a Nominal centre wavelength at 0.635, we use it as the blue channel, as it has the lowest wavelength and so is closest out of the 3 to it. We use the VIS 0.8 channel as the green channel, as it is the next step up from the blue channel, and is the next appropriate. For the red channel, we use the longest wavelength channel out of the three, IR 1.6.

In order to obtain the images from these channels, we order them from EUMetSat, through their data centre delivery service available at EUMetSat's website [EUMetSat, 2020]. This allows us to choose the channels that we require, and choose the time of day that we need the images from. For this project, as we are looking at cyclic patterns over the period of a year, so we use the most images we can get in order to have as large a data-set as possible. This means we get data from as far back as when the MSG series of satellites were operational, which is the 19th of January 2004.

Within these files however, there are multiple missing or corrupt files. Due to the Cloud free image algorithm, these have to be replaced. Due to our method of finding the cloud free image, it means that we can replace these files with simple black images that are 3712 by 3712 in size. This allows the algorithm to run smoothly. As we want to assess the vegetation coverage on the African continent alone, it meant that we can cut down the existing image to simply a rectangle around the African Continent. In addition to this, to also avoid any errors that may occur over the sea, we pass a land-mask into the program. We also pass a list of dates into the program so that the algorithm can cycle through the images.

As the images are all labelled in a pattern, it means that we can quite easily cycle through them, with the list of dates and wildcard characters to represent the rest of the string. We found that an ideal number of images to process into one cloud free image was 14, as it allowed for the African continent to be predominantly cloud free, while still retaining a relatively high sampling rate.



2.2 Cloud Free Image

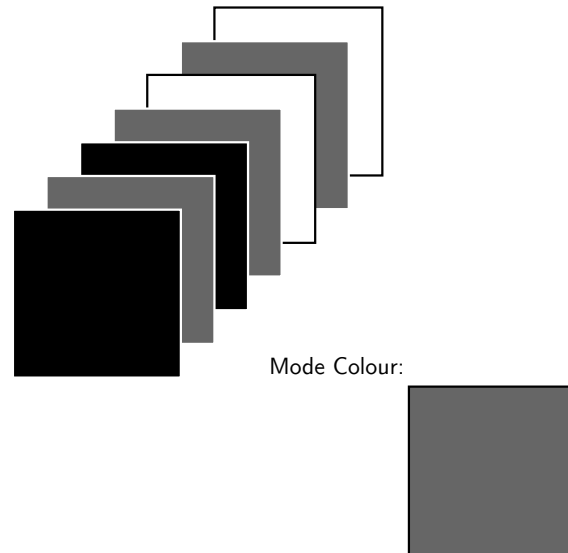
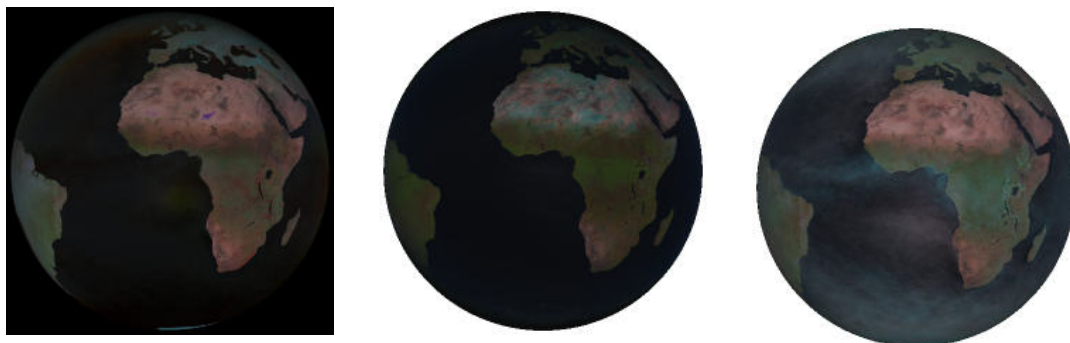


Figure 4: How the algorithm selects the ground colour for the cloud free image.

In order to properly read in the files, we need to loop through reading in 14 files from each of the red, blue and green channels. When we have these, we use the 14 individual images to create the cloud free image. As we can see in figure 4, we group the pixels together by pixel position in order to find the most common pixel in that particular position in the image. We do this for every individual pixel position within the image, which is over 5,300,000 pixels.



(a) An RGB image after having gone through a threshold and mean algorithm. (b) An RGB image after having gone through a minimum algorithm. (c) An RGB image after having gone through a mean algorithm.

Figure 5: Illustrating the effectiveness of each different cloud-removing algorithm.

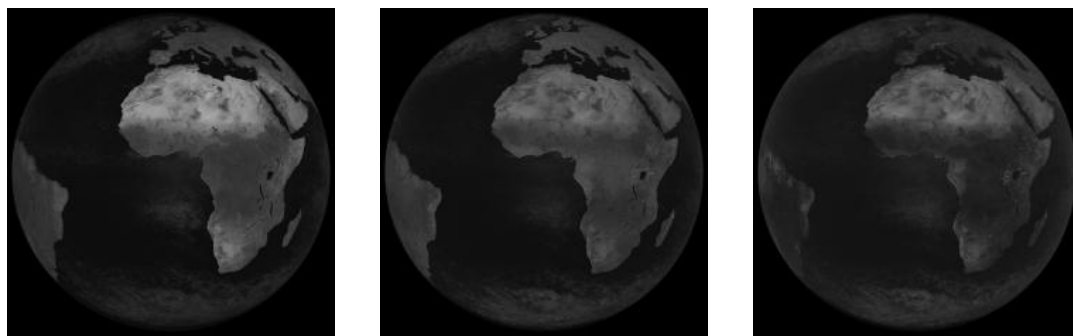
We chose to use the mode as that had the most effective results with a cloud free image, when investigating the different options available. The first one tried, shown in figure 5a, was a method involving a threshold to get rid of the bright spots (the clouds), and then averaging a selection of pixels which meet the thresholding requirements, but was too patchy to be used for this purpose. A second method shown in figure 5b, which would find the minimum of each pixel, and use that as the value, did not work either as it was too dark and miscoloured to



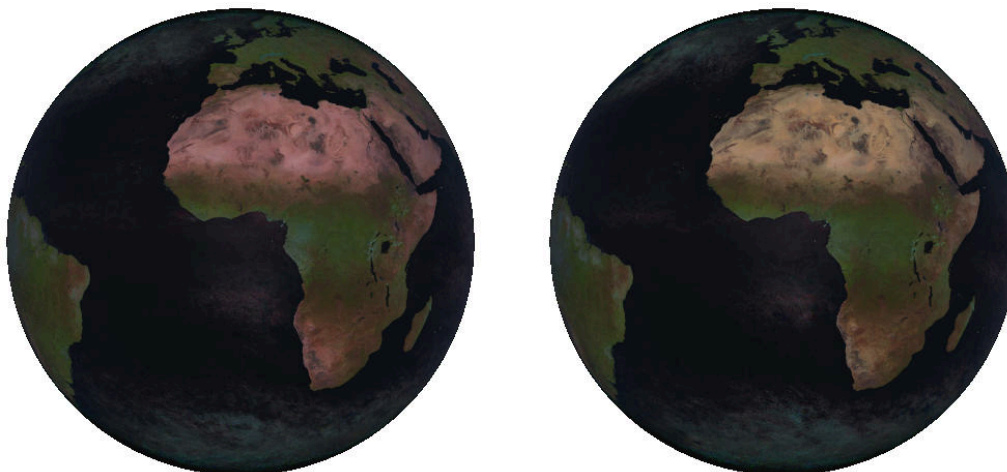
use for this purpose. The third method, shown in figure 5c, is to simply make a mean of the pixels, without thresholding. This resulted in an image with wispy "cloud" features that causes issues with the analysis of the image. When using mode however, we get the image shown in figure 6d, which is a much more accurate and full image of the earth to use for the purposes of *NDVI*.

Fortunately MATLAB can handle a mode operation through matrix operations, so it isn't necessary to set up two very large loops to solve this. After we process these images into three cloud free images, we move on to the next set of 14 images. This we do 30 more times in order to get 30 cloud free images covering just over a year of movement. Then we continue with the next 30 lots of 14 images in order to build up the full 16 year picture².

The function used to calculate the cloud free image, also estimates the standard error on each of the pixels in the new cloud free image, simply by calculating the standard deviation on the series of pixels in each position, and then dividing it by the square root of 14, the number of samples for that pixel. This, done for each pixel, produces an image of the standard error over the whole image, and for each pixel. This allows us to propagate the error forward into the *NDVI* calculations.



(a) The red channel of the cloud free image. (b) The green channel of the cloud free image. (c) The blue channel of the cloud free image.



(d) The combined non colour corrected RGB cloud free image. (e) The combined colour corrected RGB cloud free image.

Figure 6: Illustrating the combination of the three separate channels.

²This we do because of the memory (RAM) limitations of the computers we use.



In figure 6, we can see the combination of the three cloud free images into the full colour image. As defined by the International Commission on Illumination, the RGB colour space corresponds to human trichromacy [Smith and Guild, 1931], with the red wavelength at about 0.564-0.580 μm , the green wavelength at about 0.534-0.545 μm , and the blue wavelength at about 0.420-0.440 μm . As the values for the channels used do not line up with this, after obtaining the RGB image, it will need some colour correction in order to display the colours of the earth accurately. This is shown by the non colour corrected image in figure 6d, when compared to the colour corrected image in figure 6e. This colour correction however, is unnecessary when calculating $NDVI$, so is just used for the purpose of demonstrating this colour correction necessary for visual accuracy.

2.3 Normalised Difference Vegetation Index

The equation required to produce an $NDVI$ map is

$$NDVI = \frac{NIR - VIS}{NIR + VIS} \quad (1)$$

where NIR is a near-infrared channel, and VIS is a channel of visible light [NASA, 2000]. For this report, we used the red (NIR 1.6) channel, and the green (VIS 0.8) channel, for NIR and VIS , respectively. Using this, we can map where exactly the vegetation is within the African continent. First, we apply the formula to each individual pixel within a cloud free image.

This generates a matrix of the $NDVI$ values, $\begin{bmatrix} NDVI_{1,1} & \cdots & NDVI_{1,2171} \\ \vdots & \ddots & \vdots \\ NDVI_{2451,1} & \cdots & NDVI_{2451,2171} \end{bmatrix}$ where the subscript is the [row pixel position, column pixel position], which can be used to create an image, shown in figure 7a. The cloud free image that it has been generated from is shown in figure 7b.

To create a logical map based on this $NDVI$ map, we exclude all values in the $NDVI$ map below zero by setting them to 0, and set those above 0 to 1. This binary map is shown in figure 7c. For illustrative purposes, it applied to the cloud free image in figure 7b is shown in figure 7d.

In order to isolate the landmass, and exclude any error in $NDVI$ that may come from the areas of water in the cut image. To do this, we first take a general land-mask of the whole earth, as shown in figure 8a, and edit it into a land-mask of just the African continent. This is done by "white-ing out" everything but the African continent. For the purposes of this report, the African continent, or landmass, is everything from the Suez canal to the North, South, East, and West coasts, excluding Madagascar. The result of this can be seen in figure 8b. After we have applied this, we can start to count the total vegetation coverage on the landmass. This we do by summing the total number of white pixels within the land-mask boundaries, and dividing it by the total number of pixels within the land-mask boundaries.

This is done on every cloud free image, in order to get a track of the total vegetation coverage throughout the year. In order to get an estimate on the uncertainty in the $NDVI$, we average the standard error on the red and green channels, previously calculated in section 2.2, based on the standard error within the land boundaries. This gives us an estimate on the error on both channels throughout the past 16 years. By calculating the uncertainty propagation formula we can propagate this estimate into $NDVI$.





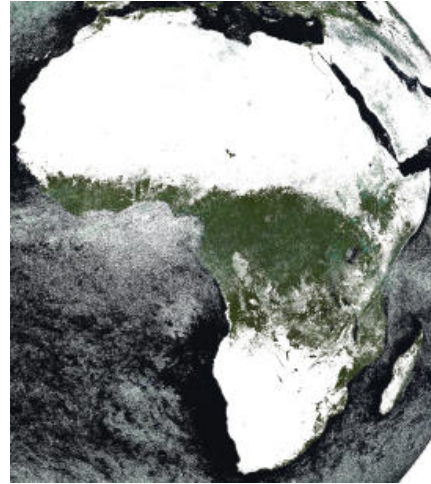
(a) The image after it has been processed to find the NDVI (Only values above 0 are shown).



(b) The cloud free image that has been processed to get that image.



(c) Image illustrating the location of healthy vegetation.



(d) The RGB image with a mask of the land-mask illustrating that it has located it correctly.

Figure 7: Diagram illustrating the effectiveness of NDVI to locate vegetation.

This can be calculated by the standard propagation of uncertainty formula, that when we have a function, $f(x, y, z, \dots)$, we can say that the uncertainty of that function can be calculated by the formula

$$\sigma_f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 \cdot \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \cdot \sigma_y^2 + \left(\frac{\partial f}{\partial z}\right)^2 \cdot \sigma_z^2 + \dots} \quad (2)$$

found in [Ku et al., 1966]. This equation in the form for the *NDVI* formula in equation 1 is

$$\sigma_{NDVI} = \sqrt{\left(\frac{\partial NDVI}{\partial NIR}\right)^2 \cdot \sigma_{NIR}^2 + \left(\frac{\partial NDVI}{\partial VIS}\right)^2 \cdot \sigma_{VIS}^2} \quad (3)$$

where σ_{NIR} is the absolute uncertainty in the near-infrared channel, σ_{VIS} is the absolute uncertainty in the visible channel, and σ_{NDVI} is the absolute uncertainty in the *NDVI*. The



partial derivatives can be calculated to be

$$\frac{\partial NDVI}{\partial NIR} = \frac{NIR + VIS - NIR + V}{(NIR + VIS)^2} = \frac{2 \cdot VIS}{(NIR + VIS)^2} \quad (4)$$

and

$$\frac{\partial NDVI}{\partial VIS} = \frac{-NIR - VIS - NIR + V}{(NIR + VIS)^2} = \frac{-2 \cdot NIR}{(NIR + VIS)^2} \quad (5)$$

Put into equation 3, this is

$$\sigma_{NDVI} = \sqrt{\frac{4NIR^2}{(NIR + VIS)^4} \cdot \sigma_{VIS}^2 + \frac{4VIS^2}{(NIR + VIS)^4} \cdot \sigma_{NIR}^2} \quad (6)$$

The proportional error in the NDVI is therefore

$$\frac{\sigma_{NDVI}}{NDVI} = \frac{2 \cdot \sqrt{\frac{\sigma_{NIR}^2 \cdot VIS^2 + NIR^2 \cdot \sigma_{VIS}^2}{(NIR + VIS)^4}}}{NDVI} \quad (7)$$

where σ_{NIR} is the absolute uncertainty in the near-infrared channel, σ_{VIS} is the absolute uncertainty in the visible channel, and σ_{NDVI} is the absolute uncertainty in the $NDVI$.

By using equation 7, we can propagate the uncertainty through to the value of $NDVI$. As the determination of if vegetation is there is based on one number of $NDVI$ (above 0), the proportional uncertainty is the same in the total vegetation coverage, as is in the $NDVI$ value. This allows us to process our values for the uncertainty in red, and the uncertainty in green, into values for the proportional uncertainty for total vegetation coverage.



(a) The land-mask used for the whole earth.



(b) The edited land-mask used to isolate the African land-mask.

In order to analyse the position and spread of the vegetation in Africa, we need to summarise each image of the binary $NDVI$ series (example shown in figure 7c). To do this we look at each row in the image, sum the number of white pixels within the land-mask, and sum the total number of pixels within the land-mask, this is shown in figure 9. This generates an intensity



value for each row, $\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_{2451} \end{bmatrix}$ where the subscript indices indicate pixel position from the top of the image, and allows us to analyse the north-south axis position and spread. If we analyse each image, we can build up a matrix, $\begin{bmatrix} I_{1,1} & \cdots & I_{1,423} \\ \vdots & \ddots & \vdots \\ I_{2451,1} & \cdots & I_{2451,423} \end{bmatrix}$ where the subscript indices are [pixel position from the top of the image, fortnights from the start of sampling].

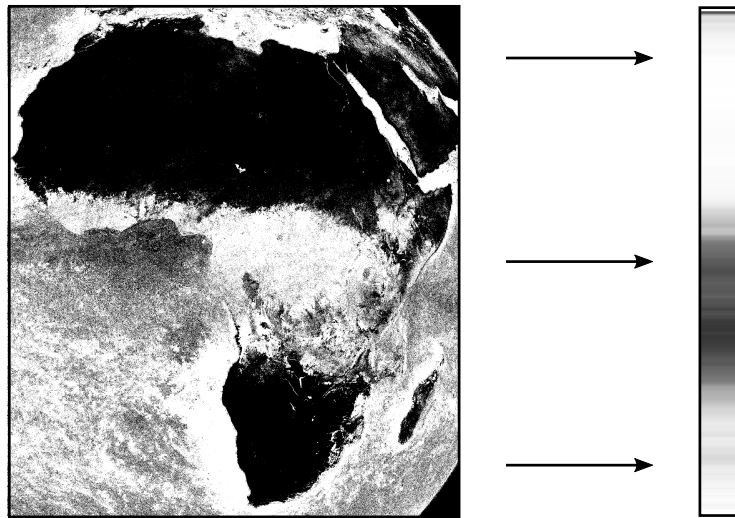


Figure 9: The arrows represent the "averaging" of each row over the landmass to get a value for the vegetation coverage on each row (darker is a higher intensity).

From this we can find the maximum position of intensity in each column. This allows us to plot a line that maps the position of the "main" part of the vegetation, which allows us to perform Fourier analysis on the position of the vegetation. To estimate the uncertainty on the maximum intensity position, we take the standard error on every column, and double it to get the 95% uncertainty boundaries.

In order to analyse the patterns in the results, we can perform Fourier analysis on the results that are, or might be periodic. By doing this to not only the main results, but their associated uncertainties and residuals as well, we can see which patterns in those main results are a result of minor variations in the pattern, and which are seasonal and cyclic patterns within the data.

3 Results

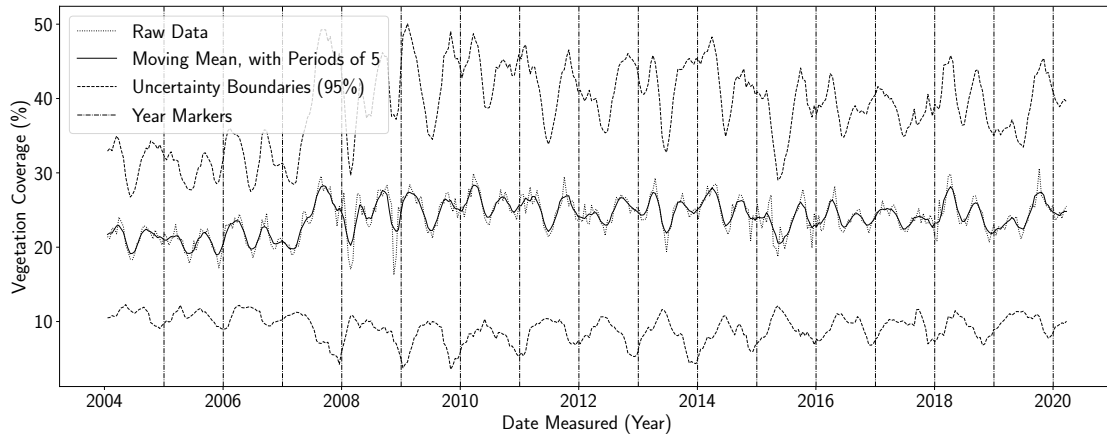


Figure 10: Total vegetation coverage over the African landmass over the last 16 years.

The graph in figure 10 illustrates the double peak in the total vegetation coverage throughout the year. This is due to the expansion of the vegetation as the vegetation moves towards the south, and expansion of the vegetation as the vegetation moves towards the north.

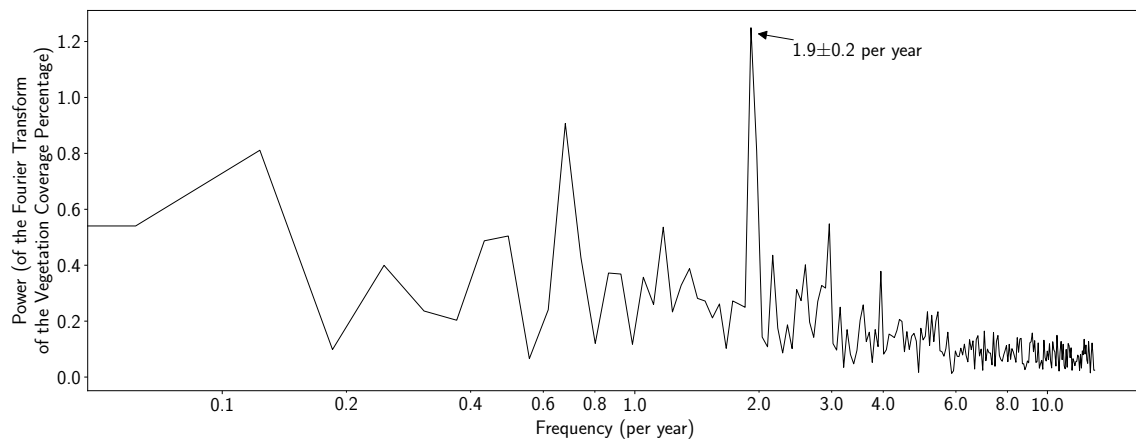


Figure 11: Fourier transform of the vegetation coverage over the African landmass over the last 16 years.

The graph in figure 11 shows the peak of the frequency of the data at 1.9 ± 0.2 , showing further the twice yearly pattern in the total vegetation coverage. We also get a smaller peak at ~ 0.7 , and another 2 smaller peaks at ~ 3 and ~ 4 .



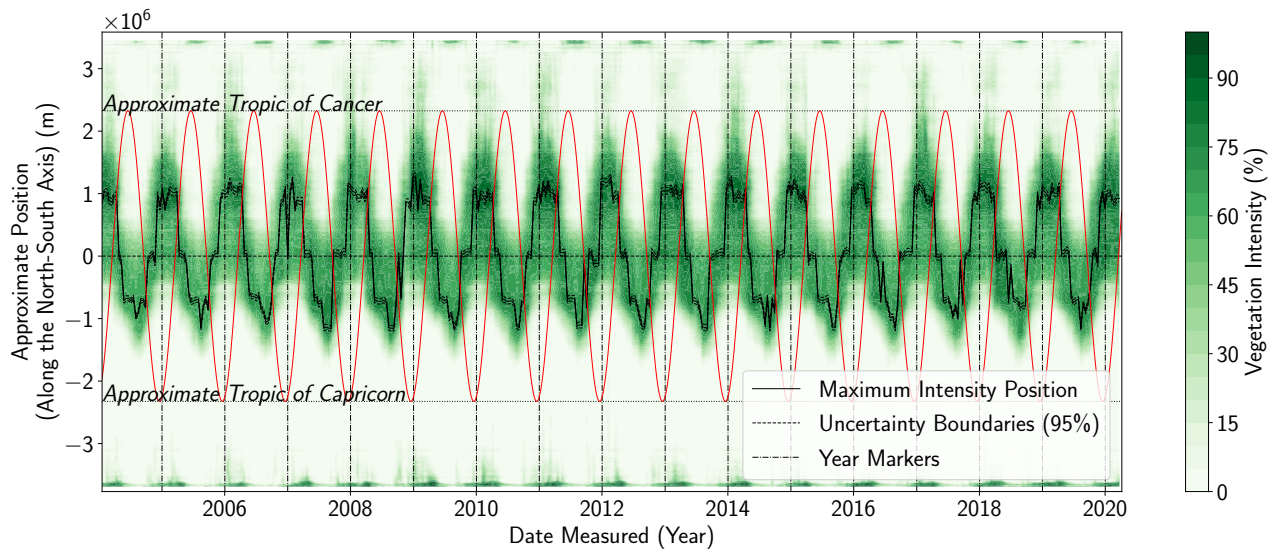


Figure 12: The position and spread, through time, of the vegetation on the North-South axis, along with the position of maximum intensity of that vegetation. The uncertainty is obscured by the thickness of the lines. The sun position throughout the year is indicated by the red line, and the centre dashed line is the position of the Equator.

In figure 12, the intensity of the green colour, corresponds to the the relative intensity of that pixel's row. It's relative to the maximum intensity of all of the rows on that image. The figure also shows that the maximum intensity position is annually periodic.

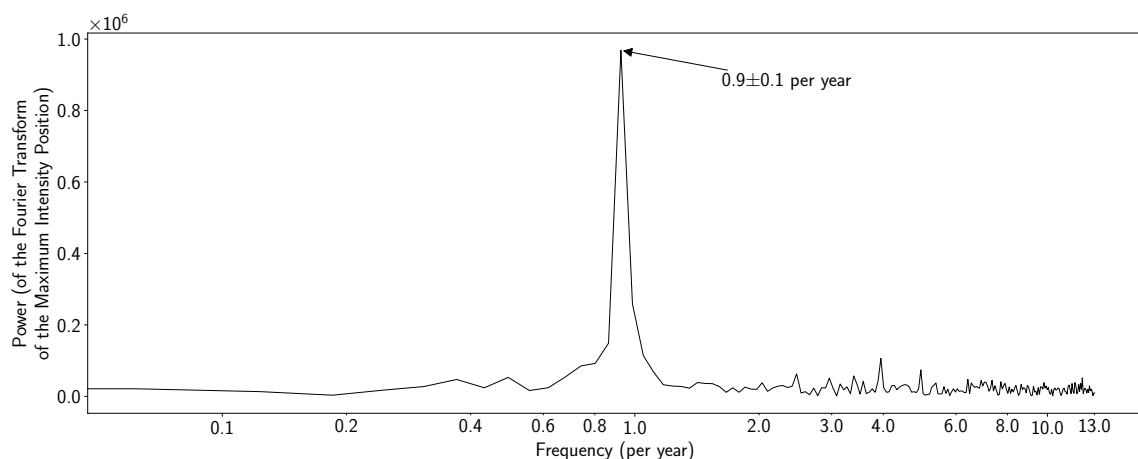


Figure 13: Fourier transform of the maximum intensity position of the vegetation.

In figure 13, we see that there is one main peak at 0.9 ± 0.1 , showing the annual pattern. There are also 2 much smaller peaks at ~ 4 and ~ 5 .



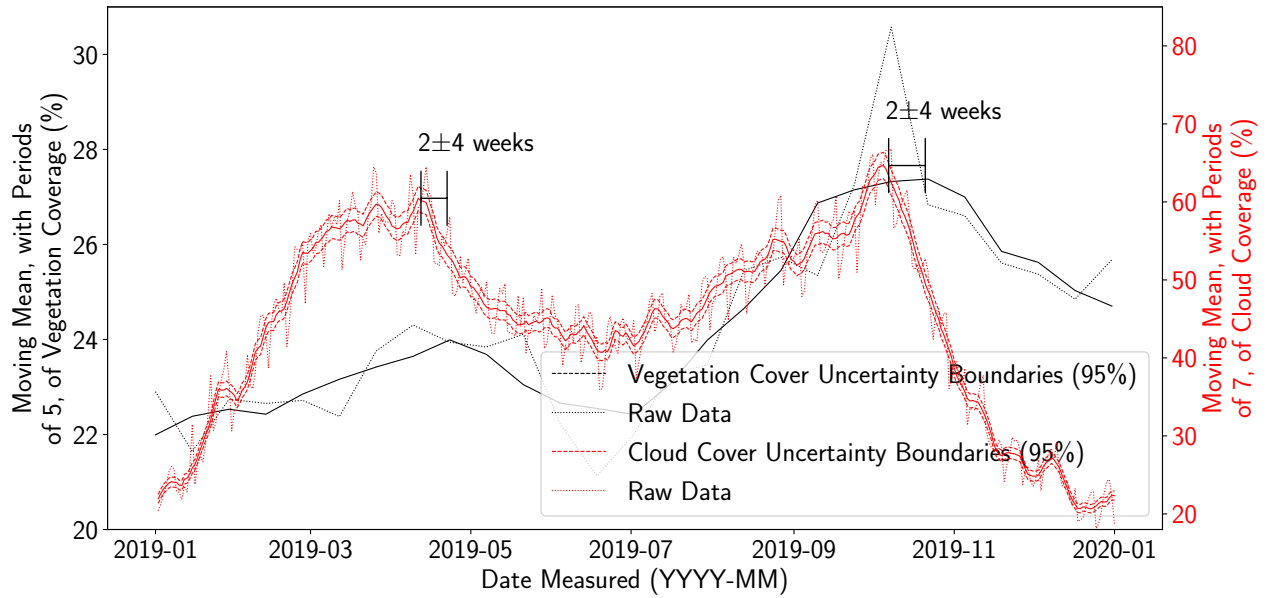


Figure 14: Cloud coverage in 2019, compared with the vegetation coverage during that year.

Figure 14 shows us the comparison between the cloud coverage in 2019 and the vegetation coverage, when we compare the peaks, we get a lead time of 2 ± 4 weeks (1.7 ± 4.2 weeks) when the vegetation is in the north of the country, and 2 ± 4 weeks (2.3 ± 4.2 weeks) when the vegetation is in the south of the country³. The uncertainty was calculated by calculating the standard error on both peaks, and combining them through equation 2 (which for a function $f = x - y$ gives $\sqrt{\sigma_x^2 + \sigma_y^2}$).

³The values in the brackets indicate the way in which the lead times are inclined towards.



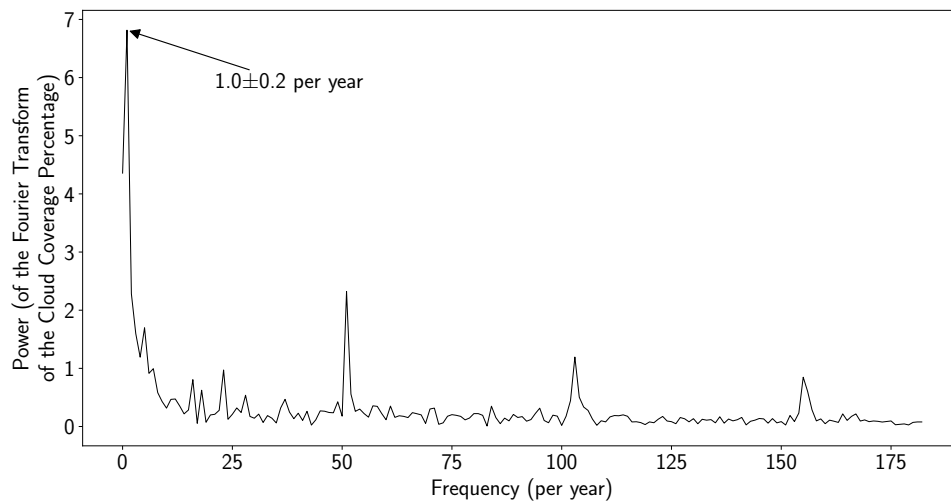


Figure 15: Fourier transform of the cloud coverage over 2019.

In figure 15, there is 1 large peak at 1.0 ± 0.2 , showing that the cloud cover has an annual pattern. There is also a peak at ~ 52 , another at ~ 104 , and another at ~ 156 .

4 Discussion of Results

The cloud coverage results came from my project partner's report, and were only created for one year. The size of the uncertainty on the lead times data is due to there being limited resolution on the total vegetation cover. A possible way to reduce this is getting cloud coverage data for more years, as more samples of lead times would mean we could take a mean of those, and therefore reduce that uncertainty. The result from figure 15 would suggest that the main pattern of the cloud cover in the area is cyclic over the period of a year, though this is an unreliable as there is not cloud coverage data for over year, which would be needed to analyse whether it is an annual pattern. By looking at the red line in figure 14, we can see that the red line also appears to have behaviour that would be described as a twice yearly pattern, which we would expect to produce a peak on figure 15 at ~ 2 . In figure 15, we see that there isn't a clear peak at ~ 2 , though there is a small peak just after 1, which is be obscured by the peak at 1.0 ± 0.2 .



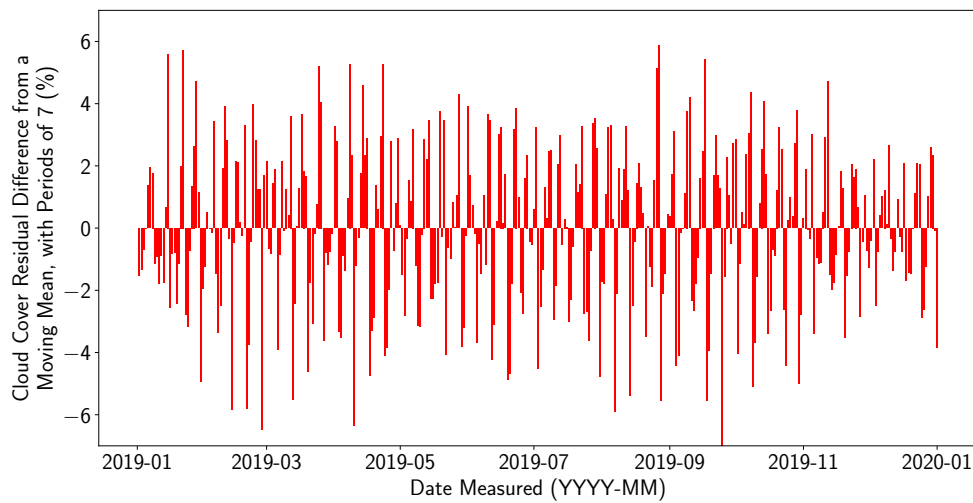


Figure 16: The residual difference of the raw data and the moving mean, with periods of 7, of the cloud coverage for 2019.

The residual difference gives us an idea of the error on the 2019 cloud coverage. If we can see a pattern in this data, it may suggest that there is a systematic error, but if there is no pattern we can say that the data simply has the expected random error. In figure 16, there appears to be a high frequency error, with the residual difference quickly flipping between positive and negative residual values. This is shown more prominently in figure 17.

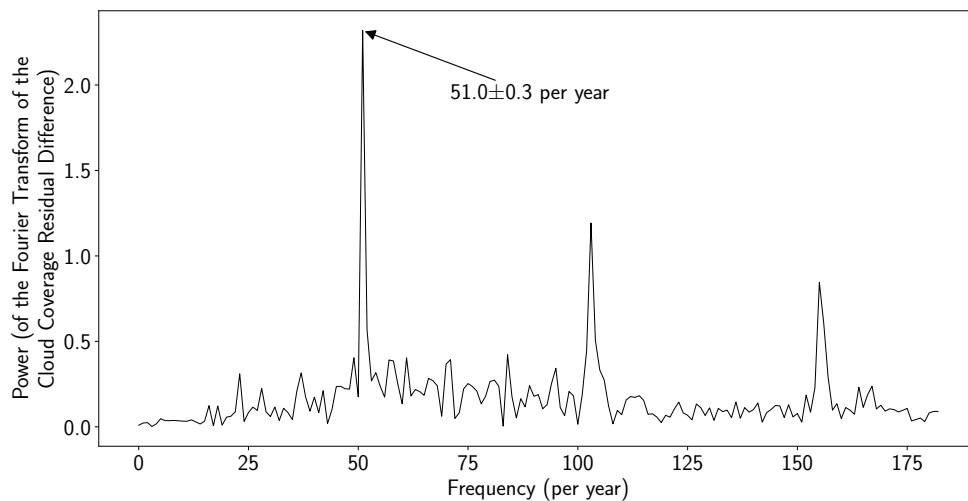


Figure 17: Fourier transform of the residual difference in cloud coverage.

The figure 17 shows us specifically the frequency of the residual difference, which at 51.0 ± 0.3 , seems to suggest that the residual difference has a weekly pattern. This pattern is also large enough to be seen in the Fourier transform of the raw cloud data in figure 15. This is odd, as there aren't any natural phenomena that occur on a weekly basis that would affect



the cloud cover. This would suggest a systematic observable error. Upon discussion with my partner, it comes to light that they had used a technique to gather the cloud data wherein they would sample it from every 7th day in a year at a time, using a cloud free image generated from those images. They would then move onto the next day. As they would generate a new cloud free image for each set, this would introduce a periodic feature into the results, as it would return to the same cloud free image every 7 days, and therefore 52 times a year. Due to this, any integer patterns in this data would show up at the multiples of ~ 52 , so that is why the peaks at ~ 104 and ~ 156 also occur. The reason the uncertainty on the cloud data isn't shown in these graphs is that it was given by my project partner to me as $\pm 2.6\%$. This would mean that a graph of this would be a replica of the black line in figure 14 but 2.6% higher.

When looking at figure 10, there are two observable levels, one which is between the start of the data and ended at the start of 2007, and then a sharp incline and another higher level between the end of 2007 and the end of the data. We could also be seeing this in figure 11, with the small peak at just after 0.1 This could be due to an error in the consistency of the calculation of NDVI, as it also occurs in the uncertainty data. This would be due to the "sectioning" and processing of the different data in parts instead of all together. Or indeed this could just be a random error in the data. However, in what is probably a coincidence, the "Great Green Wall for the Sahara and the Sahel Initiative" also started in 2007 [Nations, 2020]. This is a project in which a "wall" of trees would be planted between the Sahara and the main section of vegetation in Africa. Initially, this was done to build a "wall" which would hold back the desert, but has given way into transforming it towards a sustainable use of the land. This however is probably not the cause of the incline, as only 15% of the trees had been planted up to 2016 [Palmer, 2016], so the error in the algorithm is a more likely candidate. In the figure 10, the uncertainty is large when compared the data. This is due to the standard errors method of calculation, as we take the standard error of all of the 14 pixels used to get that one cloud free image.

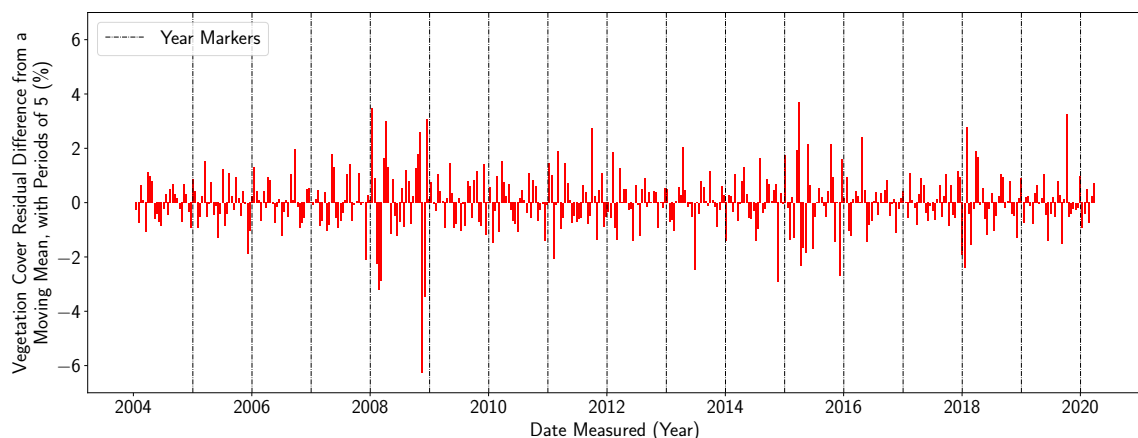


Figure 18: Residual difference between the raw data and the moving mean, with periods of 5, of the total vegetation coverage over the continent for the last 16 years.

In the figure 18, we see that the residual difference has no particular pattern or consistent height, apart from the year of 2008, which has a particularly high random error around the start and end of 2008. This is the year after the sharp incline in total vegetation coverage, so



could possibly be a result of that.

In figure 19, we can see two levels of uncertainty data, as previously mentioned when discussing the total vegetation coverage. We see the average uncertainty approximately double during 2007. This does indicate that there has been an error in the calculation of total vegetation coverage. As just the total vegetation average goes up without a change in the annual pattern, which would suggest that the error is systematic. As we are looking at the frequency of patterns, this is easily solved by carefully considering the source of the low frequency peaks in the total vegetation coverage data.

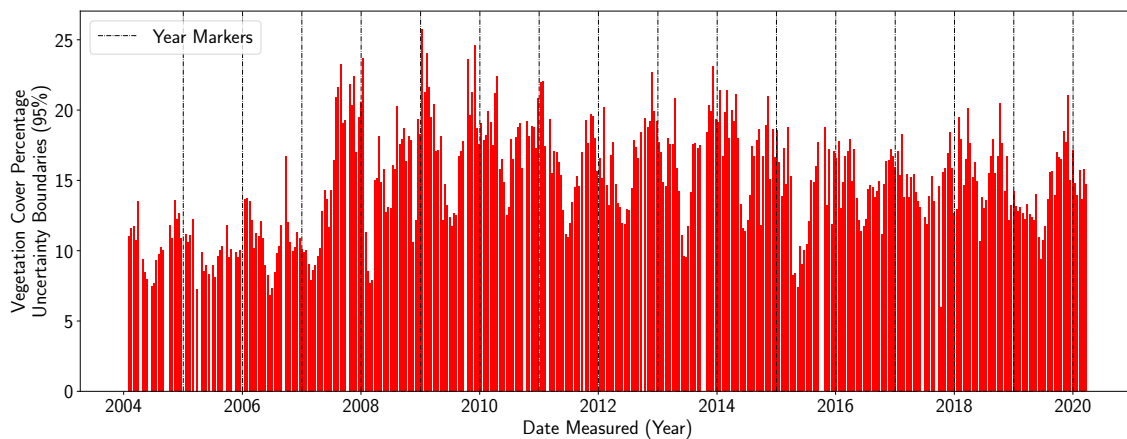


Figure 19: Estimated absolute uncertainty in the total vegetation coverage.

As previously mentioned, as we take a standard error of the 14 pixels used, we get a large standard error, as some of those pixels will have a much higher value than the value chosen. As we are without an exact value of brightness of where the ground becomes the cloud, we calculated a standard error that is much larger than the true standard error. A possible method of reducing this would be introducing the thresholding idea from the method shown in figure 5a. Using this we could exclude, approximately, the values of cloud so that we just take a standard error over that which we would say was the "ground", and get a better estimate of the uncertainty.

In order to compare the residual difference and uncertainty patterns, we look at the Fourier transform of both.



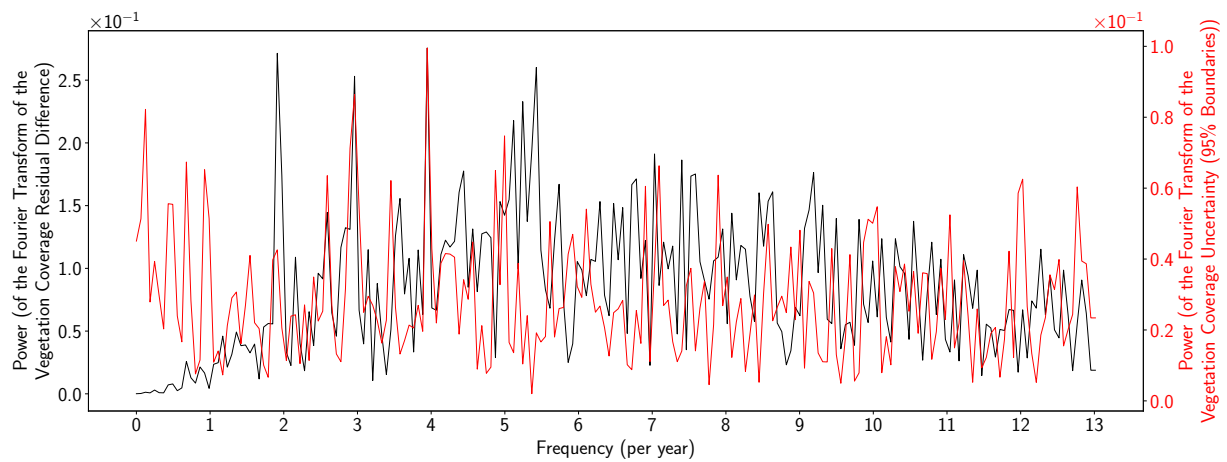


Figure 20: Fourier transform of the residual difference (black), and the estimated uncertainty (red), of the total vegetation coverage.

In figure 20, we can see that both the residual difference and uncertainty have small peaks when compared to the peaks of the Fourier transform of the total vegetation coverage. However, we can see a peak at ~ 2 for the residual difference, with a smaller size peak in the uncertainty at the same point. In the uncertainty, we also see a peak at ~ 3 and ~ 4 , with matching frequency but much bigger peaks in the the residual difference. Overall, the peaks show that the uncertainty in the total vegetation doesn't interfere with the overall pattern of the total vegetation coverage. Though, with the large spread for uncertainty, you can see that it does interfere with the "amplitude" of the total vegetation coverage.

In figure 12, the latitudinal position of the sun has been approximately plotted⁴, and you can see that, it is approximately sinusoidal with the period of a year. If we look at the track of the sun, we can see that when the sun leaves the north the vegetation tends to go towards the north, and vice versa. The thickest vegetation intensity position is mirroring the sun position, with a lower amplitude. There is also a lead time between the two, with the summer solstice in the north being before the maximum southern position of the maximum intensity position. Additionally, when the sun passes the track of the maximum intensity position, the maximum intensity position seems to stutter⁵. If we think about the vegetation as growing towards the sun, this stutter makes sense.

⁴Approximately because of the analemma of the sun.

⁵If we account for the previously mentioned lead time.





Figure 21: Image of the analemma in the afternoon of the years 1998-99, taken from outside Bell Laboratories, Murray Hill, New Jersey. Obtained from Fishburn [2014]

Imagining the vegetation as a oscillator, which has a force vector directed to the position of the sun in sky, we can quite easily see why the vegetation on the African continent behaves as it does. The stutter occurs when, accounting for the offset, the sun is directly overhead, and so the vegetation doesn't move. In addition to this, the analemma, shown in figure 21, has a large east-west position when passing the equator. Therefore, the stutter is due to the vegetation having more "encouragement" to grow east in the autumn and west in the spring. The behaviour of the vegetation is similar behaviour to phototropism, but on a much larger scale. In addition to this, the sun being directly above is not ideal growing conditions for plants. As well as this, the hot conditions causes fires in the areas below the sun, which eliminates the previously grown vegetation, and burns the grasslands [Trollope, 2010].

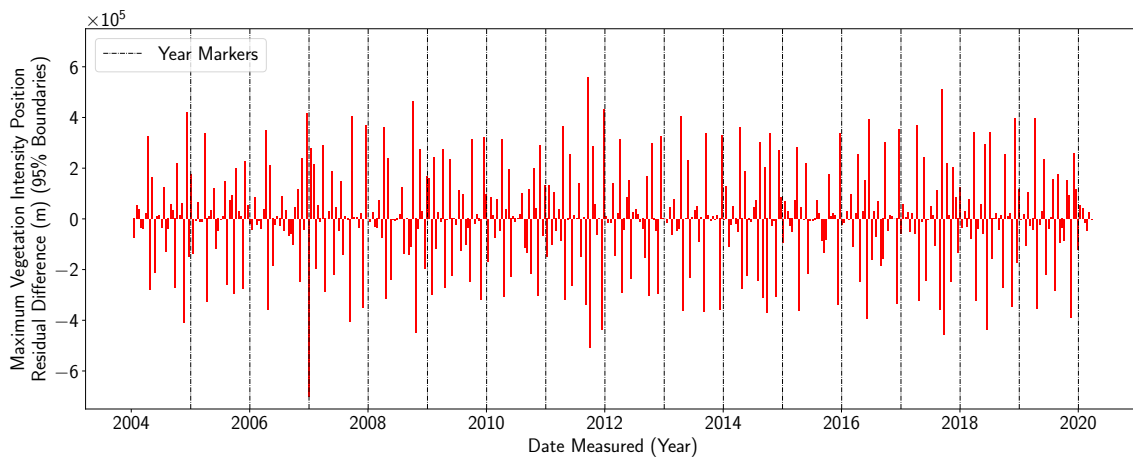


Figure 22: Residual difference between the raw data and the moving mean, with periods of 3, of the maximum intensity position over the continent for the last 16 years.

In figure 22, we can see that the residual difference is much smaller than the maximum north-south position of the maximum intensity position. The residual difference also has no clear cyclic behaviour.

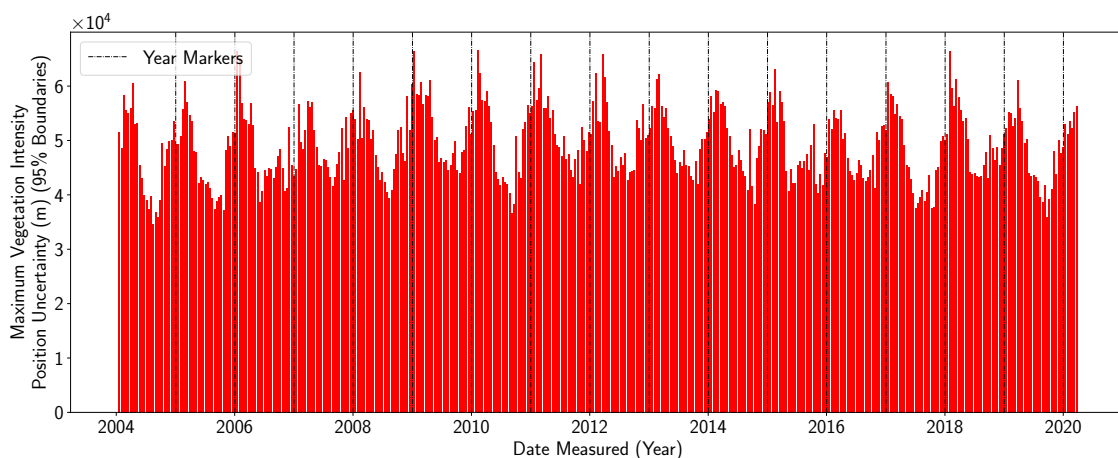


Figure 23: Estimated absolute uncertainty in the maximum intensity position.

The uncertainty in the value of the maximum intensity position is around a tenth of the value of the residual difference, and so is much smaller than that of the maximum north-south position of the maximum intensity position. The uncertainty, however, does appear to be cyclic. The uncertainty appears to be higher in the earlier half of the year, and lower in the latter half. However, instead of thinking of the uncertainty as the uncertainty, we can think of the uncertainty as the spread of the vegetation through time. From this, we can tell that in the earlier half of the year when the vegetation is in the north, the vegetation spreads more than when in the south. Looking at the NDVI maps created by the program, there seems to be more recession of vegetation from the north than there is from the south. This would indicate that the smaller spread in the south is likely caused by the larger recession of the



vegetation in the north. The larger recession in the north seems to be caused by a larger size of fires in the north compared to the south. This is due to there being larger swathes of grassland and savanna in the north when compared to the south, as the south has a forest through the centre of it. When the forest burns it sort of "gives way" to grassland, instead of to desert, whereas in the north it burns and "gives way" to the desert [Trollope, 2010]. The maximum total vegetation coverage occurs when the vegetation is transitioning from north to south in the spring, and south to north in the autumn.

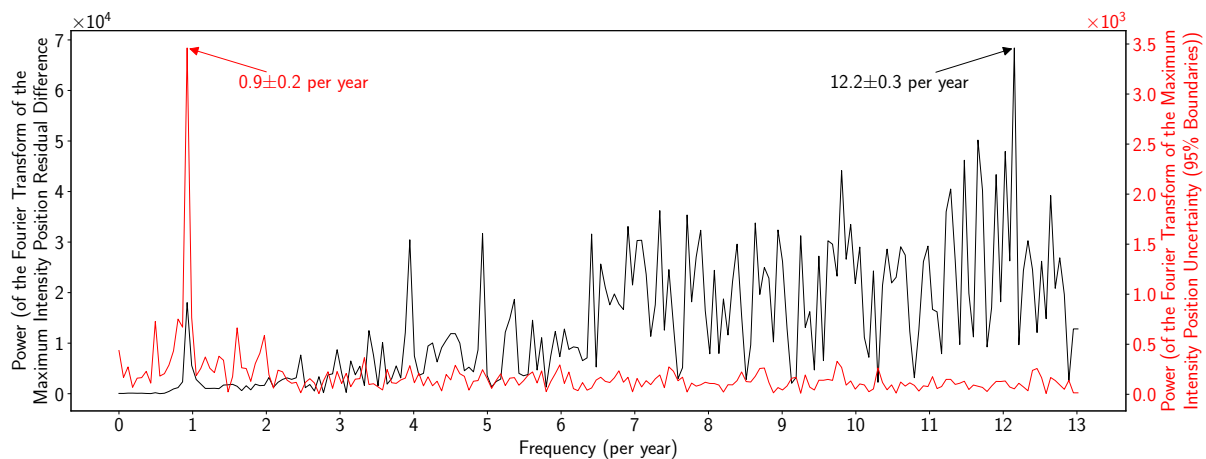


Figure 24: Fourier transform of the residual difference (black), and the estimated absolute uncertainty (red), of the maximum intensity position.

In figure 24 we can see the Fourier transform of the residual difference overlaid with the Fourier transform of the uncertainty. The residual difference has no particular pattern of note, however the highest peak in the pattern is at a frequency of 12.2 ± 0.3 per year. While 365.25 (days in a year) divided by 29.5 (days in a lunar month) is 12.4 , which is within our uncertainty, it is doubtful that what we are seeing here is this, as claims of lunar cycle influence on plants are somewhat unsubstantiated [Spiess, 2000]. The published scientific investigations also seem to suggest there is no evidence for the link [Margot, 2015]. Therefore, what we are likely looking at is a systematic cyclic error simply from the fact that there was 14 days used to create one cloud free image, which if you double and divide 365.25 by that you get roughly 13.0 , which is outside our uncertainty but is the more likely explanation. The peak does not rise far enough out of the noise to be considered statistically significant, so the lunar effect is an unlikely candidate. In the uncertainty, we can find the peak at 0.9 ± 0.2 . This indicates that the spread of the vegetation, as discussed earlier, has an annual pattern.

5 Conclusions

This report has discussed the cyclic and phenological nature of African vegetation on a continent-wide scale. We have looked at the cloud coverage of the continent, and found it to have a close relationship with the total vegetation coverage, with both having two peaks within a year. Using Fourier transforms to process the data allows us to spot the patterns in the data more easily. With the cloud data, it allowed us to see that the cloud data had an



annual pattern, with a peak in the frequency at 1.0 ± 0.2 per year. In addition to this, from figure 14, we can see it should have had a peak at around 2, but this was obscured by the peak at 1. We also saw in figure 14 that the peaks in cloud cover lead the peaks in total vegetation coverage by 2 ± 4 weeks in both the north and south, with the peaks in the first half of the year leaning towards being shorter than the latter half. The large uncertainty on this value is due to the resolution of the vegetation data. It could have been reduced through the using more cloud data for other years, as we would have more lead times for each year, which we would have been able to average and therefore reduce the uncertainty.

The total vegetation coverage allows us to see the double peak of the growth as the vegetation expands into the south during the first half of the year, and into the north in the latter half. The lower dip in the middle of the year matches with the lower spread of vegetation when it is in the south. The double dip idea matches with the Fourier transform, as we see a peak at 1.9 ± 0.2 per year. However, the uncertainty of the total vegetation coverage is very large, due to not being able to exclude the clouds when calculating uncertainty. To improve on this, we would introduce the idea of thresholding to exclude the clouds when calculating the standard error on the pixels in the channel.

The maximum intensity position allows us to see the position of the vegetation on the African continent. Using this data, we can see that the position of the vegetation is periodic. This is shown in figure 13, wherein the peak is at the frequency 0.9 ± 0.1 . We can also see that it has a relationship with the position of the sun. When the sun peaks in the northern hemisphere as a summer solstice, the vegetation reaches it's maximum southern position a few weeks later. Figure 12 also shows us that the vegetation remains stationary for a ~ 2 months at the peak and trough of it's curve, as well as stuttering on the equator. We discussed what may caused the general track of this nature, as well as specifically the stutter on the equator. This would be the vegetation following the sun, and the stutter being caused by the analemma. We also used the uncertainty to represent the spread of the vegetation, which allowed us to see that the vegetation spreads out more in the north than the south. This can be put down to the contrasting fire sizes in the north and in the south. When the vegetation is in the north, the south burns less than the north does when the vegetation is in the south. This is due to the different types of vegetation in the north and south.

All numbers with uncertainty have 95% uncertainty bounds. For all of those derived from the Fourier transform, the uncertainty was calculated by doubling the standard error of the values around the peak.

Generally, to improve this report, I'd like to have more cloud data. This would have allowed me to analyse the lead times in the peak more easily. I'd also like to have been able to analyse the cloud intensity and position through time, much like we've done with the vegetation in figure 12. Additionally, I'd have liked to do a "deep dive" in the different values of NDVI, to be able to analyse the different types of vegetation areas, to look at the recession of the vegetation in the south and north. Another systematic error that is not easily resolved is the projection of Africa, as a sphere, onto a flat surface. This causes errors as eastern Africa will take up less of the image than an equal amount of surface area would western Africa. This is the same when looking at northern-southern Africa as well. This would mean that eastern Africa would be "worth less" than western Africa in all of the summations of the vegetation coverage. This problem is not easily resolvable however, due to the impossible nature of projecting a sphere onto a plane.



References

- DMA Aminou. Msg's seviri instrument, 2002. URL http://www.esa.int/esapub/bulletin/bullet111/chapter4_bul111.pdf.
- Bernard Weymiers and Rob Oremus. Meteosat second generation, 1999. URL <http://www.esa.int/esapub/br/br153/br153.pdf>.
- John W Rouse Jr, R Hect Haas, JA Schell, and DW Deering. Monitoring the vernal advancement and retrogradation (green wave effect) of natural vegetation. *Goddard Space Flight Center*, 1973.
- Mitra Shariatinajabadi, Tiejun Wang, Andrew K Skidmore, Albertus G Toxopeus, Andrea Kölzsch, Bart A Nolet, Klaus-Michael Exo, Larry Griffin, Julia Stahl, and David Cabot. Migratory herbivorous waterfowl track satellite-derived green wave index. *PLoS One*, 9(9), 2014.
- Yali Si, Qinchuan Xin, Willem F De Boer, Peng Gong, Ronald C Ydenberg, and Herbert HT Prins. Do arctic breeding geese track or overtake a green wave during spring migration? *Scientific reports*, 5:8749, 2015.
- Richard Bischof, Leif Egil Loe, Erling L Meisingset, Barbara Zimmermann, Bram Van Moorter, and Atle Mysterud. A migratory northern ungulate in the pursuit of spring: jumping or surfing the green wave? *The American Naturalist*, 180(4):407–424, 2012.
- TH Sparks and PD Carey. The responses of species to climate over two centuries: an analysis of the marsham phenological record, 1736-1947. *Journal of Ecology*, pages 321–329, 1995.
- EUMetSat. Eumetsat's data centre, 2020. URL <https://www.eumetsat.int/website/home/Data/DataDelivery/EUMETSATDataCentre/>.
- Thomas Smith and John Guild. The cie colorimetric standards and their use. *Transactions of the optical society*, 33(3):73, 1931.
- NASA. Measuring vegetation (ndvi & evi), 2000. URL https://earthobservatory.nasa.gov/features/MeasuringVegetation/measuring_vegetation_2.php.
- Harry H Ku et al. Notes on the use of propagation of error formulas. *Journal of Research of the National Bureau of Standards*, 70(4), 1966.
- United Nations. Great green wall for the sahara and the sahel initiative, 2020. URL <http://www.fao.org/in-action/action-against-desertification/overview/great-green-wall/en/>.
- Helen Palmer. Africa's great green wall is making progress on two fronts, 2016. URL <https://www.pri.org/stories/2016-05-02/africas-great-green-wall-making-progress-two-fronts>.
- J Fishburn. Wikimedia commons: Analemma, 2014. URL https://en.wikipedia.org/wiki/File:Analemma_fishburn.tif.
- Lynne A Trollope. Fire effects and management in african grasslands and savannas. *Range and Animal Sciences and Resources Management*, 2:121–145, 2010.



Harmut Spiess. Lunar rhythms and plants. *BIODYNAMICS*, pages 19–22, 2000.

Jean-Luc Margot. Insufficient evidence of purported lunar effect on pollination in ephedra.
Journal of biological rhythms, 30(5):454–456, 2015.

